BMuSig2: Schnorr-Compatible Blind Multi-Signatures

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C Blindness of BMuSig2

Theorem C.1. *Let* GGen *and* HGen *be the group and hash generation algorithms respectively;* PKE *be a public-key encryption scheme; and* $\mathsf{NArg}[\mathsf{R}_{\mathsf{BMuSig2}}]$ *be an argument system for the relation* $\mathsf{R}_{\mathsf{BMuSig2}}$. *Then for any adversary* A *playing Game* BLD *(defined in Fig. 1) against the scheme* BMuSig2[GGen, HGen, PKE, NArg]*, there exist PPT adversary algorithms*

- $\mathsf{Z}_0, \mathsf{Z}_1$ *against the zero-knowledge of* $\mathsf{NArg}[\mathsf{R}_{\mathsf{BMuSig2}}]$*, and*
- $\mathsf{C}_0, \mathsf{C}_1$ *against the CPA-security of* PKE,

$$\mathsf{Adv}^{\mathsf{BLD}}_{\mathsf{BMuSig2},\mathsf{A}}(\lambda) \le \mathsf{Adv}^{\mathsf{ZK}}_{\mathsf{NArg}[\mathsf{R}_{\mathsf{BMuSig2}}],\mathsf{Z}_0}(\lambda) + \mathsf{Adv}^{\mathsf{ZK}}_{\mathsf{NArg}[\mathsf{R}_{\mathsf{BMuSig2}}],\mathsf{Z}_1}(\lambda) + \mathsf{Adv}^{\mathsf{CPA}}_{\mathsf{PKE},\mathsf{C}_0}(\lambda) + \mathsf{Adv}^{\mathsf{CPA}}_{\mathsf{PKE},\mathsf{C}_1}(\lambda).$$

Proof. We proceed by a sequence of games specified in Fig. 1.

Game G_0 : This is the original blindness game $\mathsf{BLD}_{\mathsf{BMuSig2}}$ from Fig. 1 (without the colored boxes). The adversary A participates as the n signers and consists of two algorithms $(\mathsf{A}_1, \mathsf{A}_2)$. First, A_1 outputs a set of verification keys $\hat{K}$ together with two messages (m_0, m_1). Next, the challenger selects a random bit $\mathsf{b} \in \{0,1\}$ and runs two concurrent signing sessions ($k = 0, 1$) for message m_b and $m_{1-\mathsf{b}}$ with A_2. The user's behaviour in these sessions is modelled through the three oracles USER_0, USER_1 and USER_2. If one or both sessions fail to output a (valid) signature, the game outputs $(\perp, \perp)$. Otherwise, if A closes both sessions successfully, the game outputs the resulting signatures (σ_0, σ_1). At the end A outputs a bit b' and wins the game if $\mathsf{b}' = \mathsf{b}$.

Game G_1 : In Game G_1, we modify the USER_1 oracle so that, instead of invoking NArg.Prove on θ, it uses NArg.SimProve to generate a simulated proof. We claim that this modification is computationally indistinguishable from G_0 due to the zero-knowledge property of $\mathsf{NArg}[\mathsf{R}_{\mathsf{BMuSig2}}]$. To prove this for $\mathsf{b} \in \{0,1\}$, we construct the adversary Z_b (Fig. 2) against the zero-knowledge game of $\mathsf{NArg}[\mathsf{R}_{\mathsf{BMuSig2}}]$, where Z_b simulates the game BLD^b for A. From the definition of the ZK game, Z_b receives as input a common reference string crs generated by NArg.Setup and MuSig2.Setup. Using this, Z_b simulates the BLD^b game for A. It answers each USER_1 query of A by invoking its Prove oracle to generate a proof π. When A_2 outputs its decision bit b', Z_b returns b' to its ZK challenger. By definition of the adversary Z_b, $\Pr\left[\mathsf{ZK}^{\mathsf{Z}_\mathsf{b},0}_{\mathsf{NArg}[\mathsf{R}_{\mathsf{BMuSig2}}]}\right]$ and $\Pr\left[\mathsf{ZK}^{\mathsf{Z}_\mathsf{b},1}_{\mathsf{NArg}[\mathsf{R}_{\mathsf{BMuSig2}}]}\right]$ denote the probabilities that Z_b wins the ZK game when the challenger provides an actual proof and a simulated proof, respectively. Therefore, we have $\Pr\left[\mathsf{ZK}^{\mathsf{Z}_\mathsf{b},0}_{\mathsf{NArg}[\mathsf{R}_{\mathsf{BMuSig2}}]}\right] = \Pr\left[\mathsf{BLD}^{\mathsf{A},\mathsf{b}}_{\mathsf{BMuSig2}}\right]$ and $\Pr\left[\mathsf{ZK}^{\mathsf{Z}_\mathsf{b},1}_{\mathsf{NArg}[\mathsf{R}_{\mathsf{BMuSig2}}]}\right] = \Pr\left[\mathsf{G}^{\mathsf{A},\mathsf{b}}_1\right]$.

We define the ZK advantage of Z_b as,

$$\begin{aligned}\mathsf{Adv}^{\mathsf{ZK}}_{\mathsf{NArg}[\mathsf{R}_{\mathsf{BMuSig2}}],\mathsf{Z}_\mathsf{b}} &:= \left|\Pr\left[\mathsf{ZK}^{\mathsf{Z}_\mathsf{b},0}_{\mathsf{NArg}[\mathsf{R}_{\mathsf{BMuSig2}}]}\right] - \Pr\left[\mathsf{ZK}^{\mathsf{Z}_\mathsf{b},1}_{\mathsf{NArg}[\mathsf{R}_{\mathsf{BMuSig2}}]}\right]\right| \\ &= \left|\Pr\left[\mathsf{BLD}^{\mathsf{A},\mathsf{b}}_{\mathsf{BMuSig2}}\right] - \Pr\left[\mathsf{G}^{\mathsf{A},\mathsf{b}}_1\right]\right|.\end{aligned}$$

Applying the triangle inequality, this implies

$$\begin{aligned}\mathsf{Adv}^{\mathsf{BLD}}_{\mathsf{BMuSig2},\mathsf{A}} &= \left|\Pr\left[\mathsf{BLD}^{\mathsf{A},1}_{\mathsf{BMuSig2}}\right] - \Pr\left[\mathsf{BLD}^{\mathsf{A},0}_{\mathsf{BMuSig2}}\right]\right| \\ &= \Big|\Pr[\mathsf{BLD}^{\mathsf{A},1}_{\mathsf{BMuSig2}}] - \Pr[\mathsf{G}^{\mathsf{A},1}_1] + \Pr[\mathsf{G}^{\mathsf{A},1}_1] - \Pr[\mathsf{BLD}^{\mathsf{A},0}_{\mathsf{BMuSig2}}] \\ &\quad + \Pr[\mathsf{G}^{\mathsf{A},0}_1] - \Pr[\mathsf{G}^{\mathsf{A},0}_1]\Big| \\ &\le \mathsf{Adv}^{\mathsf{ZK}}_{\mathsf{NArg}[\mathsf{R}_{\mathsf{BMuSig2}}],\mathsf{Z}_1} + \left|\Pr[\mathsf{G}^{\mathsf{A},1}_1] - \Pr[\mathsf{G}^{\mathsf{A},0}_1]\right| + \mathsf{Adv}^{\mathsf{ZK}}_{\mathsf{NArg}[\mathsf{R}_{\mathsf{BMuSig2}}],\mathsf{Z}_0}.\end{aligned} \tag{1}$$

Game G_2 : In Game G_2 we modify the USER_0 oracle to encrypt a arbitrary fixed message $\bar{m}$ and $\alpha = 0, \beta = 0$ instead of the chosen message. We claim that the game G_2 is computationally indistinguishable from the game G_1 due to the CPA-security of PKE. To prove this, for $\mathsf{b} \in \{0,1\}$, we define an adversary algorithm C_b (Fig. 3) that plays the CPA game for the PKE scheme. From the definition of CPA, C_b receives the encryption key ek, extracts the security parameter 1^λ, and generates (sp, crs) to simulate $\mathsf{G}^{\mathsf{A},\mathsf{b}}_1$. On a call to $\mathsf{USER}_0(k)$, C_b sets $M_0 := (\bar{m}, 0, 0)$, $M_1 := (m_{k\oplus\mathsf{b}}, \alpha_k, \beta_k)$, and queries its encryption oracle on (M_0, M_1), and forwards the received ciphertext C_k to A_2. (Note: C_b does not know the randomness used to generate C_k but this is sufficient since proofs in USER_1 are simulated and do not require the witness). By definition of C_b, $\Pr[\mathsf{CPA}^{\mathsf{C}_\mathsf{b},0}_{\mathsf{PKE}}]$ and $\Pr[\mathsf{CPA}^{\mathsf{C}_\mathsf{b},1}_{\mathsf{PKE}}]$ denote the probabilities that C_b wins the CPA game when the challenger encrypts an actual message and a fixed message, respectively. Therefore we have $\Pr[\mathsf{CPA}^{\mathsf{C}_\mathsf{b},0}_{\mathsf{PKE}}] = \Pr[\mathsf{G}^{\mathsf{A},\mathsf{b}}_2]$ and $\Pr[\mathsf{CPA}^{\mathsf{C}_\mathsf{b},1}_{\mathsf{PKE}}] = \Pr[\mathsf{G}^{\mathsf{A},\mathsf{b}}_1]$. We define the CPA advantage of C_b as

$$\mathsf{Adv}^{\mathsf{CPA}}_{\mathsf{PKE},\mathsf{C}_\mathsf{b}} := \left|\Pr[\mathsf{CPA}^{\mathsf{C}_\mathsf{b},1}_{\mathsf{PKE}}] - \Pr[\mathsf{CPA}^{\mathsf{C}_\mathsf{b},0}_{\mathsf{PKE}}]\right| = \left|\Pr[\mathsf{G}^{\mathsf{A},\mathsf{b}}_1] - \Pr[\mathsf{G}^{\mathsf{A},\mathsf{b}}_2]\right|.$$

Applying the triangle inequality, this implies

$$\begin{aligned}\left|\Pr[\mathsf{G}^{\mathsf{A},1}_1] - \Pr[\mathsf{G}^{\mathsf{A},0}_1]\right| &= \Big|\Pr[\mathsf{G}^{\mathsf{A},1}_1] - \Pr[\mathsf{G}^{\mathsf{A},1}_2] + \Pr[\mathsf{G}^{\mathsf{A},1}_2] - \Pr[\mathsf{G}^{\mathsf{A},0}_2] \\ &\quad + \Pr[\mathsf{G}^{\mathsf{A},0}_2] - \Pr[\mathsf{G}^{\mathsf{A},0}_1]\Big| \\ &\le \mathsf{Adv}^{\mathsf{CPA}}_{\mathsf{PKE},\mathsf{C}_1} + \left|\Pr[\mathsf{G}^{\mathsf{A},1}_2] - \Pr[\mathsf{G}^{\mathsf{A},0}_2]\right| + \mathsf{Adv}^{\mathsf{CPA}}_{\mathsf{PKE},\mathsf{C}_0}.\end{aligned} \tag{2}$$

Proving blindness from game G_2*:* The signer's view after the successful completion of the two signing sessions $k \in \{0,1\}$ consists of i) parameters (crs, ek), ii) signatures with their corresponding

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Game $\mathsf{BLD}^{\mathsf{A,b}}_{\mathsf{BMuSig2}}(\lambda)$, G_1, G_2

$sp := (q, \mathbb{G}, g, \mathsf{H}_{\mathsf{agg}}, \mathsf{H}_{\mathsf{non}}, \mathsf{H}_{\mathsf{sig}}) \leftarrow \mathsf{MuSig2.Setup}(1^\lambda)$
$(crs, \tau) \leftarrow \mathsf{NArg.Setup}(sp)$
$(ek, dk) \leftarrow \mathsf{PKE.KeyGen}(1^\lambda)$
$(m_0, m_1, \hat{K}) \leftarrow \mathsf{A}_1(crs)$
$(sess_0, sess_1) := (init, init)$
// Sample randomness used by $\mathsf{USER}_0(\cdot)$
$(\alpha_0, \alpha_1, \beta_0, \beta_1) \xleftarrow{\$} \mathbb{Z}_q^4$
$(\rho_0, \rho_1) \xleftarrow{\$} \mathcal{R}^2_{ek}$
$\mathsf{b}' \leftarrow \mathsf{A}_2^{\mathsf{USER}_0, \mathsf{USER}_1, \mathsf{USER}_2}(\,)$
return $(\mathsf{b}' = \mathsf{b})$

Oracle $\mathsf{USER}_0(k)$:

if $sess_k \neq init$ **then return** $\perp$
$sess_k := open$
$M := (m_{k\oplus\mathsf{b}}, \alpha_k, \beta_k)$
// encrypt fixed message vector instead
$M := (\bar{m}, 0, 0)$
$C_k := \mathsf{PKE.Enc}(ek, M; \rho_k)$
return C_k

Oracle $\mathsf{USER}_1(k, \hat{K}, \{pst_j\}_{j\in[n]})$:

$(R^{(k)}_{j,1}, \ldots, R^{(k)}_{j,\nu}) :\subseteq pst_j$ (for $j \in [n]$)
if $sess_k \neq open$: **return** $\perp$
$sess_k := await$
$\forall t \in [\nu] : R^{(k)}_t := \prod_{j=1}^n R^{(k)}_{j,t}$
$X \leftarrow \mathsf{BMuSig2.KeyAgg}(\hat{K})$
$b_k := \mathsf{H}_{\mathsf{non}}(X,\ R^{(k)}_1, \ldots, R^{(k)}_\nu, m_{k\oplus\mathsf{b}})$
$R_k := \prod_{t=1}^\nu R^{(k)}_t{}^{b_k^{t-1}}$
$R'_{k\oplus\mathsf{b}} := R_k \cdot g^{\alpha_k} X^{\beta_k}$
$c_k := \big(\mathsf{H}_{\mathsf{sig}}(X, R'_{k\oplus\mathsf{b}}, m_{k\oplus\mathsf{b}}) + \beta_k\big) \bmod q$
$\theta := (X, R^{(k)}_1, \ldots, R^{(k)}_\nu, b_k, R_k, c_k, C_k, ek)$
$w := (m_{k\oplus\mathsf{b}}, \alpha_k, \beta_k, \rho_k)$
$\pi \leftarrow \mathsf{NArg.Prove}(crs, \theta, w)$
// Simulate the proof instead
$\pi \leftarrow \mathsf{NArg.SimProve}(crs, \tau, \theta)$
return $(R^{(k)}_1, \ldots, R^{(k)}_\nu, b_k, c_k, \pi)$

Oracle $\mathsf{USER}_2(k, \{\sigma_j\}_{j\in[n]})$:

$(R'_{k\oplus\mathsf{b}}, s^{(k)}_j) :\subseteq \sigma_j$ (for $j \in [n]$)
if $sess_k \neq await$: **return** $\perp$
$sess_k := closed$
$s_k := \sum_{j=1}^n s^{(k)}_j \bmod q$
if $g^s = R_k \cdot X^{c_k}$:
  $\sigma_{k\oplus\mathsf{b}} := \big(R'_{k\oplus\mathsf{b}}, (s_k + \alpha_k) \bmod q\big)$
else $\sigma_{k\oplus\mathsf{b}} := \perp$
if $sess_0 = sess_1 = closed$:
  if $\sigma_0 = \perp \vee \sigma_1 = \perp$:
    return $(\perp, \perp)$
  return (σ_0, σ_1)
return ε

Figure 1: The blindness game for BMuSig2 and hybrid games used in the proof of Theorem C.1. Game G_0 (without colored boxes) is the original BLD game, Game G_1 additionally includes the green box, and Game G_2 further includes the orange box.

Game $\mathsf{Z}^{\mathsf{PROVE}}_{\mathsf{b}}(crs)$

$(q, \mathbb{G}, g, \mathsf{H}_{\mathsf{agg}}, \mathsf{H}_{\mathsf{non}}, \mathsf{H}_{\mathsf{sig}}) :\subseteq crs$
$(ek, dk) \leftarrow \mathsf{PKE.KeyGen}(1^\lambda)$
$(m_0, m_1, \hat{K}) \leftarrow \mathsf{A}_1(crs, ek)$
$(sess_0, sess_1) := (init, init)$
// Sample randomness used by $\mathsf{USER}_0(\cdot)$
$(\alpha_0, \alpha_1, \beta_0, \beta_1) \xleftarrow{\$} \mathbb{Z}_q^4$
$(\rho_0, \rho_1) \xleftarrow{\$} \mathcal{R}^2_{ek}$
$\mathsf{b}' \leftarrow \mathsf{A}_2^{\mathsf{USER}_0, \mathsf{USER}_1, \mathsf{USER}_2}(\,)$
return b'

Oracle $\mathsf{USER}_1(k, \hat{K}, \{pst_j\}_{j\in[n]})$:

$(R^{(k)}_{j,1}, R^{(k)}_{j,2}) :\subseteq pst_j$ (for $j \in [n]$)
if $sess_k \neq open$: **return** $\perp$
$sess_k := await$
$R^{(k)}_1 := \prod_{j=1}^n R^{(k)}_{j,1}, R^{(k)}_2 := \prod_{j=1}^n R^{(k)}_{j,2}$
$X \leftarrow \mathsf{BMuSig2.KeyAgg}(\hat{K})$
$b_k := \mathsf{H}_{\mathsf{non}}(X,\ (R^{(k)}_1, R^{(k)}_2), m_{k\oplus\mathsf{b}})$
$R_k := R^{(k)}_1 \cdot R^{(k)}_2{}^{b_k}$
$R'_{k\oplus\mathsf{b}} := R_k \cdot g^{\alpha_k} X^{\beta_k}$
$c_k := \big(\mathsf{H}_{\mathsf{sig}}(X, R'_{k\oplus\mathsf{b}}, m_{k\oplus\mathsf{b}}) + \beta_k\big) \bmod q$
$\theta := (X, R^{(k)}_1, R^{(k)}_2, b_k, R_k, c_k, C_k, ek)$;
$w := (m_{k\oplus\mathsf{b}}, \alpha_k, \beta_k, \rho_k)$
$\pi \leftarrow \mathsf{PROVE}(\theta, w)$
return
$(R^{(k)}_1, R^{(k)}_2, b_k, c_k, \pi)$

Figure 2: The game Z_{b} playing against zero-knowledge of the $\mathsf{NArg}[\mathsf{R}_{\mathsf{BMuSig2}}]$. The USER_0, USER_2 oracles simulated to A_2 are same as G_0.

Game $\mathsf{C}^{\mathsf{ENC}}_{\mathsf{b}}(ek)$

$1^\lambda :\subseteq ek$
$sp := (q, \mathbb{G}, g, \mathsf{H}_{\mathsf{agg}}, \mathsf{H}_{\mathsf{non}}, \mathsf{H}_{\mathsf{sig}}) \leftarrow \mathsf{MuSig2.Setup}(1^\lambda)$
$(crs, \tau) \leftarrow \mathsf{NArg.Setup}(sp)$
$(m_0, m_1, \hat{K}) \leftarrow \mathsf{A}_1(crs, ek)$
$(sess_0, sess_1) := (init, init)$
// Sample randomness used by $\mathsf{USER}_0(\cdot)$
$(\alpha_0, \alpha_1, \beta_0, \beta_1) \xleftarrow{\$} \mathbb{Z}_q^4$
$(\rho_0, \rho_1) \xleftarrow{\$} \mathcal{R}^2_{ek}$
$\mathsf{b}' \leftarrow \mathsf{A}_2^{\mathsf{USER}_0, \mathsf{USER}_1, \mathsf{USER}_2}(\,)$
return b'

Oracle $\mathsf{USER}_0(k)$:

if $sess_k \neq init$ **then return** $\perp$
$sess_k := open$
$M_0 := (\bar{m}, 0, 0)$
$M_1 := (m_{k\oplus\mathsf{b}}, \alpha_k, \beta_k)$
$C_k := \mathsf{ENC}(M_0, M_1)$
return C_k

Figure 3: The game C_{b} playing against CPA security of PKE. The USER_1, USER_2 oracles simulated to A_2 are same as G_1.

messages, $(m_0, \sigma_0 := (R'_0, s'_0))$ and $(m_1, \sigma_1 := (R'_1, s'_1))$, and *iii*) transcripts T_k corresponding to the n signers, where T_k is defined as

$$T_k := \big(C_k, (R^{(k)}_{j,1}, \ldots, R^{(k)}_{j,\nu}), (R^{(k)}_1, \ldots, R^{(k)}_\nu), R_k, b_k, c_k, \pi, \{s^{(k)}_j\}\big),$$

where $k \in \{0, 1\}$ and $j \in [n]$. Since the ciphertext C_k is an encryption of a fixed message using randomness ρ_k, and π is a simulated proof, neither of them reveals any information about the bit b. Therefore, we exclude these components from our analysis of transcripts and user states.

We define the view of adversary (signer) A as

$$V_{k,i}(\mathsf{A}) := \big(T_k, m_i, \sigma_i\big), \quad \text{for } k, i \in \{0, 1\}.$$

First, we show that for all $k, i \in \{0, 1\}$, the view $V_{k,i}(\mathsf{A})$ determines a valid user state $st := (\alpha_{k,i}, \beta_{k,i})$. This st [1] represents the internal values the user would have used if session k corresponds to message m_i. We can construct it as follows,

$$\begin{aligned} \alpha_{k,i} &:= s'_i - s_k \bmod q, \text{ where } s_k = \sum_{j \in [n]} s_j^{(k)}, \\ \beta_{k,i} &:= c_k - \mathsf{H}_{\mathsf{sig}}(X, R'_i, m_i) \bmod q. \end{aligned} \tag{3}$$

Next, we have to show that these values are uniformly distributed. The uniformity of $\alpha_{k,i}$ follows from the uniformity of s'_i which is computed as $(s_i + \alpha_i)$, where α_i is the real blinding factor used during the experiment. Similarly, the uniformity of $\beta_{k,i}$ follows from c_k, which consists of the real blinding factor β_k in the experiment.

Now, we can define $R'_{k,i},\ s_{k,i},\ c_{k,i}$ as

$$R'_{k,i} := \prod_{t=1}^{\nu} R_t^{(k)\, b_k^{t-1}} \cdot g^{\alpha_{k,i}} X^{\beta_{k,i}}, s_{k,i} := s_k - \alpha_{k,i}, c_{k,i} := \mathsf{H}_{\mathsf{sig}}(X, R'_{k,i}, m_i).$$

Finally, we show that for all $k, i \in \{0, 1\}$, the user state st defines a valid signature (R'_i, s'_i) and hash c'_i. From the validity of signatures $\sigma_i = (R'_i, s'_i)$, we have

$$g^{s'_i} = R'_i \cdot X^{c'_i} \Rightarrow R'_i = g^{s'_i} \cdot X^{-c'_i}, \tag{4}$$

where $c'_i = \mathsf{H}_{\mathsf{sig}}(X, R'_i, m_i)$. Now substitute $s'_i = \alpha_{k,i} + s_k$ and $c'_i = c_k - \beta_{k,i}$ from Eqn. 3 in Eqn. 4 to obtain

$$R'_i = g^{\alpha_{k,i} + s_k} \cdot X^{-c_k + \beta_{k,i}}. \tag{5}$$

We assume A closes both signing sessions $k \in \{0, 1\}$ successfully, and in this case, T_k is a valid transcript. Therefore we have,

$$g^{s_k} = R_k \cdot X^{c_k}, \text{ where } R_k = \prod_{t=1}^{\nu} R_t^{(k)\, b_k^{t-1}}. \tag{6}$$

We substitute the above value for g^{s_k} in Eqn.5 to obtain

$$\begin{aligned} R'_i &= g^{\alpha_{k,i}} \cdot (R_k \cdot X^{c_k}) \cdot X^{-c_k + \beta_{k,i}} = g^{\alpha_{k,i}} \cdot R_k \cdot X^{\beta_{k,i}} \\ &= R_k \cdot g^{\alpha_{k,i}} X^{\beta_{k,i}} = \prod_{t=1}^{\nu} R_t^{(k)\, b_k^{t-1}} \cdot g^{\alpha_{k,i}} X^{\beta_{k,i}} = R'_{k,i}. \end{aligned}$$

We can see that R'_i is determined by the user state. Consequently it holds that $s'_i = s_k - \alpha_{k,i} \bmod q$ and $c'_i = \mathsf{H}_{\mathsf{sig}}(X, R'_{k,i}, m_i)$. This means both session transcripts T_0 and T_1 can define the signature (R'_i, s'_i). Hence, the advantage in distinguishing G_2 with b = 0 from G_2 with b = 1 is

$$\left| \Pr[\mathrm{G}_2^{\mathsf{A},1}] - \Pr[\mathrm{G}_2^{\mathsf{A},0}] \right| = 0. \tag{7}$$

Eqn. 7, together with Eqns. 1 and 2 concludes the proof.

□

[1] Note that the user state is determined via $st := (\alpha_{k,i}, \beta_{k,i})$ because all the other values on the user side are fixed (given the transcripts and the message-signature pair).